

# Research Synthesis & Upgrade Plan

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## AI-Driven Marp/Markdown Presentation Workflow

# Agenda

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## Today's Topics

### 1. Design Best Practices

Mayer's principles, typography, layout

### 2. Color, Accessibility & Data Viz

WCAG, palettes, chart frameworks

### 3. Narrative & Anti-Patterns

SCQA, slide count, F1-F8

### 4. AI Tool Landscape

6 tools surveyed, key gap found

### 5. Open-Source Pipelines

sunbigfly, Akxan, DeepSlide

### 6. Vision QA & Upgrade Roadmap

VLM contracts, 4-phase plan

# Design Best Practices

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- **Cognitive Load Theory** – Mayer's 12 principles, validated across 200+ experiments
- **Coherence** – every element must serve the message, nothing extraneous
- **Signaling** – use contrast and bold for key takeaways, never font size alone
- **Segmenting** – max 5 +/- 2 items per slide, progressive disclosure
- **Pre-training** – define key terminology upfront in title and outline



# Typography & Visual Hierarchy

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- **7-level scale:** 48pt > 40pt > 28pt > 22pt > 18pt > 14pt > 12pt
- **Line height:** body 1.5x, headings 1.1x (unitless CSS)
- **Hard rules:** body  $\geq 14\text{px}$  and  $\leq 24\text{px}$ , max 3 font families
- **7 bento templates:** full-bleed, 2-col, 3-col, 4-card, 6-card, center



# Color, Accessibility & Data Viz

4.5:1

WCAG AA body contrast

3:1

Large text minimum

5

Max colors per slide

**Principle:** Pair icons, patterns, or labels with color. Never rely on color alone to convey information (WCAG 2.1 SC 1.4.1).



## Palette

Base > Surface > Primary > Accent > Text



## Best Charts

Bar, line, scatter, stacked bar



## Avoid

Pie >3 segments, 3D, dense tables

# Narrative Structure & Failure Modes

## SCQA

Situation > Complication > Question > Answer

1:2

Pacing ratio – 1 slide per 2 minutes, 15 min = 7-8 slides

**F2**

### Overfill

Reduce font or count

**F6**

### Contrast Failure

Adjust colors per WCAG

**F8**

### Narrative Disconnect

Restructure with SCQA

# AI Presentation Tool Landscape

| Tool         | Layout Engine           | Quality Mechanism          |
|--------------|-------------------------|----------------------------|
| Gamma.app    | Template-constrained    | Human review, no visual QA |
| Beautiful.ai | Constraint satisfaction | Template design rules only |
| Tome         | Narrative-first         | No pixel-level QA          |
| Canva Magic  | Score heuristics        | Layout balance ranking     |
| MS Copilot   | Template-constrained    | Template guardrails        |
| Pitch        | Design system           | Human review only          |

**Key gap:** No major tool uses vision feedback to refine slide appearance

# Open-Source Agentic Pipelines

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## **sunbigfly/ppt-agent-skills**

Multi-agent state machine, 4 subagents, JSON contract architecture, Pillow pixel QA, stateless resume



## **Akxan/ppt-agent-skill**

26 production styles, 14 typography rules, 18 chart types (SVG), 8 failure modes with fix protocols



## **DeepSlide (arXiv)**

Time-budgeted planner, Markov style inheritance, dual-scoreboard evaluation (artifact + delivery)

# Common Gaps & Best Format

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## VLM Gap

No open-source tool uses vision-language models for semantic slide QA. Analysis is purely pixel-statistical.

### BEST FORMAT

HTML/CSS as primary target, convert to Marp markdown

### DEEPSLIDE INNOVATION

Logical-chain planner with time budgets, style inheritance, dual-scoreboard evaluation

# Vision QA & Style Contracts

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- **Stage A:** per-slide visual audit – balance, readability, color
- **Stage B:** deck-wide narrative audit – arc, progression, pacing
- Use VLM for **semantic** checks, CSS/DOM for **quantitative**
- **theme-contract.json:** full design system encoding per deck



# Upgrade Roadmap

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## Phase 1 – Foundation

Style contract system,  
CSS/DOM inspection QA,  
density contract validator

WEEK 1

## Phase 2 – Pipeline

5-stage pipeline shell, per-  
slide JSON contracts,  
stateless resume support

WEEK 2

## Phase 3 – Quality

VLM QA pass with structured  
prompts, feedback loop,  
chart MCP tool

WEEK 3

## Phase 4 – Polish

Icon search MCP tool,  
narrative arc planning,  
integration testing

WEEK 4